*.ops

Purpose

The six `.ops` files declared in `type input_ops` are parameter databases for the management operations that `management.sch` and the `lum.dtl` decision tables can invoke. Each file is a named list of parameter sets that other files reference by `name`.

All six readers follow the same structure: skip two header lines, count records, allocate, rewind, skip two, read one record per row into the corresponding `*_db` array.

Source

- Readers: `src/mgt_read_harvops.f90`, `src/mgt_read_grazeops.f90`, `src/mgt_read_irrops.f90`, `src/mgt_read_chemapp.f90`, `src/mgt_read_fireops.f90`, `src/mgt_read_sweepops.f90`.
- Type definitions: `src/mgt_operations_module.f90`.

File format

For every file:

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3 and following: one record per operation. List-directed read. The reader counts records on a first pass, allocates the database array, rewinds, and reads the values.

The "Common" first column is the operation name; other files reference it.

harv.ops

Reader: `mgt_read_harvops.f90` → `harvop_db` (`type harvest_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	typ	char(40)	“”	none	`grain`, `biomass`, `residue`, `tree`, or `tuber`
3	hi_ovr	real	0	none	harvest index override at harvest
4	eff	real	0	fraction	harvest efficiency. Fraction of harvested yield removed; the remainder is left as residue
5	bm_min	real	0	kg/ha	minimum biomass required to allow harvest

graze.ops

Reader: `mgt_read_grazeops.f90` → `grazeop_db` (`type grazing_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	fertnm	char(40)	“”	none	manure name pointing into `fertilizer.frt`
3	manure_id	int	0	none	resolved id of `fertnm` in `fertilizer.frt`
4	eat	real	0	(kg/ha)/day	dry biomass removed by grazing per day
5	tramp	real	0	(kg/ha)/day	dry biomass removed by trampling per day
6	manure	real	0	(kg/ha)/day	dry manure deposited per day
7	biomin	real	0	kg/ha	minimum plant biomass to allow grazing

irr.ops

Reader: `mgt_read_irrops.f90` → `irrop_db` (`type irrigation_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	amt_mm	real	25.4	mm	application amount
3	eff	real	0	none	in-field application efficiency
4	surq	real	0	fraction	surface runoff ratio
5	dep_mm	real	0	mm	depth of application for subsurface irrigation
6	salt	real	0	mg/kg	total salt concentration in irrigation water
7	no3	real	0	mg/kg	nitrate concentration
8	po4	real	0	mg/kg	phosphate concentration

chem_app.ops

Reader: `mgt_read_chemapp.f90` → `chemapp_db` (`type chemical_application_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	form	char(40)	“”	none	`solid` or `liquid`
3	op_typ	char(40)	“”	none	`spread`, `spray`, `inject`, or `direct`
4	app_eff	real	0	none	application efficiency
5	foliar_eff	real	0	none	foliar efficiency
6	inject_dep	real	0	mm	injection depth
7	surf_frac	real	0	fraction	fraction in upper 10 mm of soil
8	drift_pot	real	0	none	drift potential
9	aerial_unif	real	0	none	aerial uniformity

fire.ops

Reader: `mgt_read_fireops.f90` → `fire_db` (`type fire_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	cn2_upd	real	0	none	change applied to SCS curve number II value
3	fr_burn	real	0	fraction	fraction of cover burned

sweep.ops

Reader: `mgt_read_sweepops.f90` → `sweepop_db` (`type streetsweep_operation`).

#	Field	Type	Default	Units	Description
1	name	char(40)	“”	none	operation name
2	eff	real	0	none	removal efficiency of the sweeping operation
3	fr_curb	real	0	none	availability factor: fraction of curb length that is sweepable

Example

`refdata/Ames_sub1/harv.ops`:

```
harv.ops written by SWAT+ editor v2.3.3 on 2023-10-25
name                  typ                hi_ovr        eff        bm_min
harv_grain            grain              0.0           1.0        0.0
harv_residue          residue            0.0           1.0        0.0
```

The other five files follow the same shape with their own column sets.

Related files

- `landuse.lum` and `management.sch` reference operations by `name`.
- `*.dtl` decision tables can fire operations by `name`.
- `fertilizer.frt` provides the `manure_id` resolution target for `graze.ops`.